Revision — 1.1.3 Input, Output and Storage Devices

OCR H446 — one-page revision summary

Must know

- **Input** devices send data into a system (mouse, keyboard, capacitive touchscreen, barcode scanner, microphone, sensor); **output** devices act on the world (monitor, printer, speakers, **actuator**).
- There is **no single "right" device** — justify a choice against the scenario: environment, user/accessibility, accuracy, speed, cost, durability.
- **Printers**: **laser** = fast, high-volume, low cost per page, sharp text (offices); **inkjet** = cheaper to buy, good occasional colour/photos, low volume (home); **3D** = builds objects layer by layer.
- **Secondary storage** is **non-volatile** — it retains data without power.
- **Magnetic** (HDD, tape): magnetised patterns on a spinning platter/tape read by a R/W head — huge capacity, low £/GB, but slow and fragile (moving parts). Tape = **serial access**, ideal for cheap archive/backup.
- **Flash/SSD** (SSD, USB, SD): EEPROM traps electrons in cells, **no moving parts** — fast, silent, low power, durable, portable; costlier/GB and finite write cycles (uses **wear levelling**).
- **Optical** (CD/DVD/Blu-ray): a laser detects **pits and lands** via reflected light — very cheap and portable, but low capacity, slow and easily scratched.
- **RAM** = volatile, read/write, holds running programs/data; **ROM** = non-volatile, normally read-only, holds bootstrap/BIOS firmware.
- **Virtual memory** = when RAM is full, the OS moves less-used **pages** to a swap/paging file on secondary storage; excessive paging causes **disk thrashing**. **Virtual (cloud) storage** = remote servers accessed over a network (access anywhere, scalable, off-site backup; needs internet).

Key terms

Term	Definition
Volatile	Loses its contents when power is removed (e.g. RAM)
Non-volatile	Retains contents without power (secondary storage, ROM)
RAM	Volatile read/write memory holding currently running programs and data
ROM	Non-volatile, normally read-only memory holding bootstrap/BIOS firmware
Flash / SSD	Solid-state storage using EEPROM cells with no moving parts
Serial access	Data read/written in sequence (e.g. magnetic tape)
Virtual memory	Use of secondary storage as extra RAM when physical RAM is full
Disk thrashing	Excessive paging where the system spends most time swapping pages, not working

Worked example / how to — recommending a storage device

Scenario: portable storage for large video files, moved between sites, frequently rewritten, needing fast transfer and durability.

1. **List the requirements** — portable, large capacity, fast transfer, frequently rewritten, durable.
2. **Recommend a device** — an external **flash/SSD** (or USB flash drive).
3. **Justify against each requirement** — small/light and portable; **fast** transfer suits large video files; easily **rewritable**; **durable** (no moving parts to damage when carried).
4. **Compare/reject alternatives** — an HDD is cheaper per GB but has fragile moving parts; optical media is slow and low capacity — so flash best fits the stated needs.

Exam tips

- Distinguish **input vs output** carefully — an actuator is an **output**, a sensor/microphone is an input.
- For "choose a device/storage" questions, **link each characteristic to the scenario** and **compare** ("whereas…"); listing generic pros/cons without applying them caps marks at AO1.
- Keep the three "virtuals" separate: **virtual memory** (secondary storage as extra RAM) vs **virtual/cloud storage** (remote servers) — read which one the question means.
- Be precise on mechanisms: magnetic = **magnetised patterns**, optical = **pits and lands** read by laser, flash = **electrons trapped in cells, no moving parts**.